Physiological Measurement

PAPER

Fine-tuning ECG-FM for Chagas screening once a recording-length shortcut is controlled: internal evidence and one hidden-cohort score

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Keywords: Chagas disease, electrocardiogram, foundation model, fine-tuning, shortcut learning, PhysioNet Challenge, generalisation

Abstract

Objective. Our entry to the George B. Moody PhysioNet Challenge 2025 detected Chagas disease from the 12-lead electrocardiogram (ECG) with the ECG-FM foundation model kept frozen. We asked whether fine-tuning the pretrained encoder would do better, and how to judge that on the public data. *Approach.* From the entry, we added one change at a time: soft labels, augmentation, two-stage fine-tuning with layer-wise learning-rate decay, unpadded windows of ECG-FM's pretraining length, a hard-negative ranking loss and seed ensembling. Models were compared on a patient-grouped held-out split, within recording length, under decision rules written before each round. *Main results.* Unfreezing the transformer layers improved on the stage-1 head it started from (TPR@5% 0.375 vs 0.343), but we could not show that it beats a frozen encoder trained to convergence; our re-created frozen entry came within noise of the fine-tuned model. Pretraining-length windows and the ranking loss together removed the length cue. That cue is a property of the data: in CODE-15%, the largest training source, short records are 12.7 times as often positive as long ones, record length alone reaches an AUROC of 0.677, and every model we trained used it. Scored by the organisers on the hidden test cohorts, the revised entry raised our score from 0.245 to 0.289. *Significance.* ECG-FM tolerates a short second stage of fine-tuning and gains from it. The larger lesson is about evaluation: on these data, none of our changes could be judged until the comparison was made within recording length.

1 Introduction

Chagas disease, caused by *Trypanosoma cruzi*, affects millions of people in Latin America and is often silent for decades before it produces cardiomyopathy. Serological testing is the diagnostic standard but capacity is limited, so an inexpensive test that ranks people for serology is valuable. The 12-lead ECG is a natural candidate, because Chagas cardiomyopathy leaves conduction and rhythm abnormalities such as right bundle branch block and left anterior fascicular block (Jidling et al., 2023).

The George B. Moody PhysioNet Challenge 2025 asked teams to rank ECGs by the probability of Chagas disease (Reyna et al., 2025; Goldberger et al., 2000). The score is the fraction of positive cases found among the 5% of records with the highest predicted probability (TPR@5%). Training data came from three public sources with labels of very different quality: CODE-15% with self-reported labels (Ribeiro et al., 2020), SaMi-Trop with serologically confirmed positives (Cardoso et al., 2016), and PTB-XL with presumed negatives (Wagner et al., 2020). Models were scored on three hidden, serologically labelled Brazilian cohorts.

Several strong entries started from an ECG encoder pretrained on other data, and they did not agree on whether to fine-tune it. The winning team found full fine-tuning of its vision-transformer foundation model better than a head on the frozen model, with a cross-validated score of 0.490 against 0.381 (Van Santvliet et al., 2025); another team found a frozen encoder better on the hidden data (Wang et al., 2025). With labels as noisy as these, fine-tuning might overwrite what

Table 1. Records by source and length. CODE-15% records are 7.34 s or 10.24 s long; the handful shorter than 5 s fall in the first group.

		Training split		Held-out split		Prevalence
Source	Record length	Records	Positive	Records	Positive	(%)
CODE-15%	< 7.5 s	163,655	4,977	40,867	1,259	3.05
	7.5–9.9 s	5,226	6	1,273	0	0.09
	≥ 9.9 s	105,299	240	26,435	77	0.24
SaMi-Trop	< 7.5 s	564	564	147	147	100.00
	7.5–9.9 s	74	74	16	16	100.00
	≥ 9.9 s	667	667	163	163	100.00
PTB-XL	≥ 9.9 s	17,400	0	4,399	0	0.00

pretraining taught the encoder. Or it might be the only way to reach what the disease leaves in the signal.

Our entry kept the ECG-FM encoder (McKeen et al., 2025) frozen and trained a small head on its output, with age and sex concatenated to the pooled feature (Nguyen et al., 2025a). It scored 0.245 on the hidden test set, eighth of the ranked teams. Internally, unfreezing the encoder had cost us 0.08 of score, and our internal and hidden scores disagreed: the model that scored best on our own split, trained without CODE-15%, scored 0.081 on the hidden validation cohort.

We were not alone: across the ranked teams the median fell from 0.279 on the hidden validation cohort to 0.236 and 0.100 on two of the test cohorts, the best test mean was 0.323 (Reyna et al., 2025), and one team watched a variant gain 0.20 locally and lose 0.11 on hidden validation (Wang et al., 2025). Disagreement this consistent suggests that models learned a feature that predicts the label in the training data for reasons unrelated to the disease, a *shortcut* (Geirhos et al., 2020) (in medical imaging, shortcuts have been traced to the hospital and the acquisition device; Zech et al., 2018; DeGrave et al., 2021). Here every SaMi-Trop record is positive and every PTB-XL record negative, and the winning team warned that the original sampling frequency and the mains frequency could act as cues of this kind (Van Santvliet et al., 2025). A shortcut also gets in the way of the fine-tuning question, because a pooled comparison rewards whichever model exploits it best.

We went back to the entry to ask how much more the pretrained encoder had to give. We fine-tuned its transformer layers in two stages, fed it windows of the length it was pretrained on, added a ranking loss and averaged seeds, one change at a time, and judged each change by a rule written down before the runs. Making those judgements meant first dealing with a property of the public data that we have not seen described: inside CODE-15%, the length of a recording predicts its Chagas label. Section 2.1 states it, and from there on we compare models within recording length. A revised entry chosen this way was scored by the organisers on the hidden cohorts (section 3.6).

2 Methods

2.1 Data and evaluation

We used the public Challenge training data: CODE-15% (400 Hz), SaMi-Trop (400 Hz) and PTB-XL (500 Hz), 366,185 records after conversion to the Challenge format. CODE-15% labels are self-reported, SaMi-Trop labels are serologically confirmed and all positive, and PTB-XL records are presumed negative because of their European origin (Reyna et al., 2025). CODE-15% contains repeated examinations of the same patient, so we grouped records by patient and held out 20% of the groups, stratified by source and label. The held-out set has 73,300 records and 1,662 positives, at the natural prevalence of 2.27%; this matters because TPR@5% cannot exceed 0.05 divided by the prevalence. The other 292,885 records form the training split (table 1).

One property of these data shapes everything that follows. CODE-15% recordings come in two lengths, and the label depends on which: 3.05% of the 204,522 records of 7.34 s are positive, against 0.24% of the 131,734 records of 10.24 s, a ratio of 12.7 (figure 1(a)). Of the CODE-15% positives, 95.1% are short records. Used alone, record length ranks the held-out CODE-15% records with an AUROC of 0.677. The hidden cohorts are all about 10 s long (Reyna et al., 2025). Whatever a model learns from length, it cannot use there.

We therefore score models within recording length. *S7* is the held-out CODE-15% records shorter than 7.5 s (40,867 records, 1,259 positives); neither length nor source can help there, and it has the most positives. *S10* is the CODE-15% records of at least 9.9 s (26,435 records, 77 positives), the format of the hidden cohorts. *S10+* adds to the negatives of S10 every held-out positive of that length; 163 of its 240 positives are SaMi-Trop records, which buys power at the price of a source cue. View *A*, all held-out records, rewards both cues and is reported for continuity

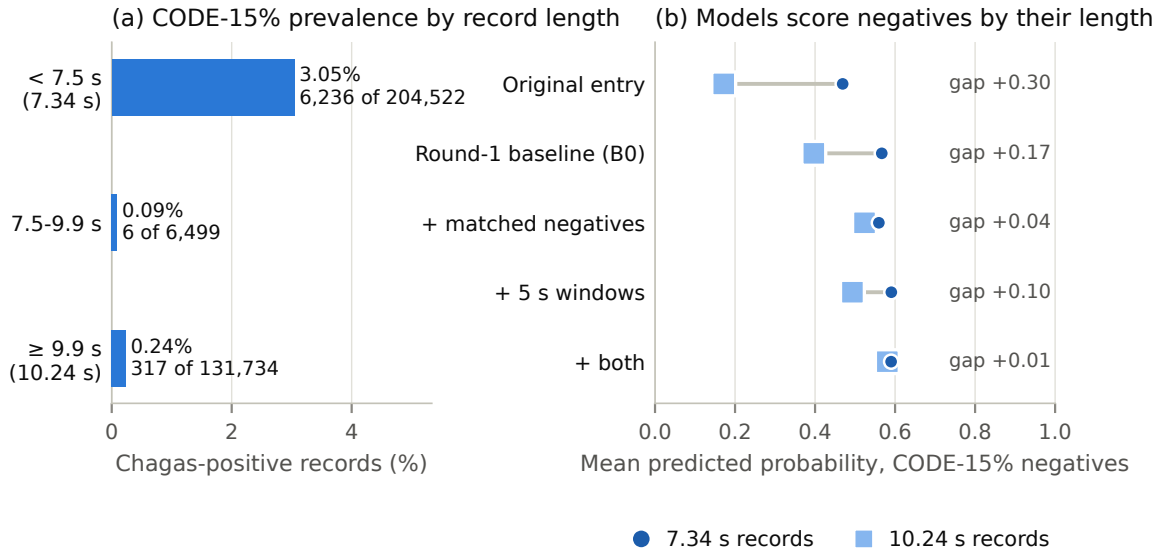


Figure 1. The length shortcut. (a) Prevalence of the Chagas label in CODE-15% by record length. (b) Mean predicted probability of held-out CODE-15% negatives of 7.34 s (circles) and 10.24 s (squares), by model; means over seeds where there are several.

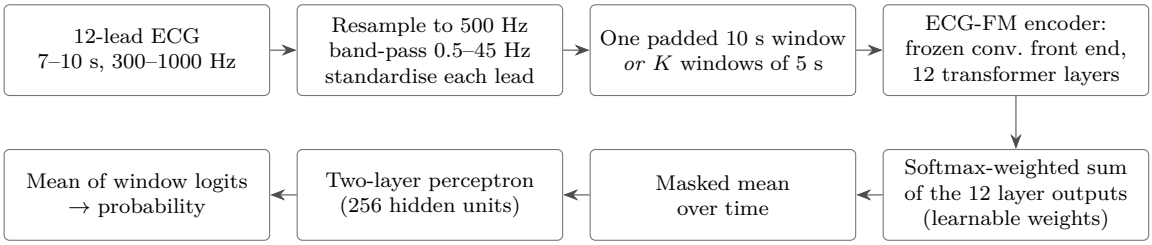


Figure 2. The model. Stage 1 trains the layer weights and the perceptron on the frozen encoder; stage 2 also fine-tunes the 12 transformer layers. The fixed-window variant averages the logits of $K = \lceil T/5\,\mathrm{s} \rceil$ evenly spaced 5 s windows of a record of length T.

(views B and C are described in supplementary section S7). Three diagnostics go with every comparison: the *length-cue gap*, the mean predicted probability of short CODE-15% negatives minus that of long ones; the mean predicted probability of PTB-XL records, all negative; and the *source gap*, the mean probability of SaMi-Trop positives minus that of CODE-15% positives.

The primary metric is TPR@5%; we also report the AUROC, which does not depend on prevalence. Confidence intervals are 95% percentile intervals from 1,000 bootstrap resamples of the test records (Efron and Tibshirani, 1993). Two configurations are compared with a paired bootstrap that resamples the same records for both, and we report the standard error (SE) of the difference. Unless stated otherwise a configuration was trained with three seeds and we give the mean and standard deviation across them; a *seed ensemble* averages the three models' probabilities.

2.2 Model and training

Figure 2 shows the model. ECG-FM is a convolutional front end followed by 12 transformer layers of width 768, pretrained on 5 s segments sampled at 500 Hz with each lead standardised (McKeen et al., 2025); as in our Challenge entry, we used the released checkpoint that its authors had further fine-tuned on the diagnostic labels of MIMIC-IV-ECG (Gow et al., 2023). Leads were put in a fixed order, resampled to 500 Hz, band-pass filtered between 0.5 and 45 Hz with a second-order, zero-phase Butterworth filter (the design of Hong et al., 2025), notch-filtered at 50 and 60 Hz, and standardised over the record. The 45 Hz cut-off and the notches are ours. In the baseline the first 10 s of a record form one window, and shorter records are zero-padded at the end.

The classifier wrapper released with ECG-FM detaches the encoder output from the computation graph, so we read the output of each transformer layer with a forward hook. The 12 outputs are combined with learnable softmax weights, initialised to be uniform, and averaged over time; padded positions are left out of the average and masked in the encoder's attention. (In our Challenge entry the mask was dead code, and the zero padding of a 7.34 s record was averaged in

as if it were signal.) A head of dropout, a linear layer to 256 units, a ReLU, dropout and a linear layer gives one logit. None of this is in the released ECG-FM code: the hooks, the layer weights, the pooling and the head are ours.

Training has two stages, with the schedule and learning rates of the winning entry (Van Santvliet et al., 2025). Stage 1 trains only the layer weights, the demographic network where there is one and the head, for 2,000 optimiser steps at 2×10^{-4} with the encoder frozen. Stage 2 unfreezes the 12 transformer layers but not the convolutional front end, as in the encoder's own fine-tuning recipe; layer k is trained at $2 \times 10^{-5} \cdot 0.85^{12-k}$ (Howard and Ruder, 2018) and the head at 10^{-4}, with a cosine schedule and 5% warm-up. The batch is 16, accumulated over four batches, in 16-bit mixed precision, with weight decay 0.01 and gradient clipping at 1.0. We kept the checkpoint with the best validation TPR@5% and stopped after three validations without improvement.

The loss is binary cross-entropy with a positive-class weight of 5, and positives were sampled five times as often as negatives. Targets are soft for the self-reported source: 0.8 and 0.2 for CODE-15% positives and negatives, 1 for SaMi-Trop and 0 for PTB-XL (Van Santvliet et al., 2025). For CODE-15% patients with several examinations the target is $0.51\,y + 0.49\,n_+/n$, where y is the record's label and n_+/n the fraction of the patient's examinations labelled positive (Hong et al., 2025). All positives were used, and negatives were subsampled so that a run saw 100,000 records in total. Most augmentations follow Hong et al. (2025) and Van Santvliet et al. (2025): resampling to and from another rate, device-like filtering, baseline wander, Gaussian noise, amplitude scaling, time warping and random masking. The mains-interference and crop-and-shift augmentations are our own, from the Challenge code. Every run went through the unmodified Challenge entry scripts, on an NVIDIA Titan RTX with memory capped at 14.5 GiB to mimic the Challenge's 16 GB card (timings in supplementary section S6).

2.3 Experimental design

The work ran in three rounds, each judged by a rule fixed in writing before its runs (table 2; full texts, with dates, in supplementary section S1). Round 1 built the model up from our Challenge entry one change at a time. Step 0 re-creates the entry: frozen encoder, age and sex concatenated to the pooled feature, hard labels and class-balanced sampling. Step 1 adds pipeline corrections (the working padding mask, and a vectorised band-pass filter in place of the original signal cleaner) and early stopping; step 2 the soft labels and class weighting; step 3 augmentation. Step 4 replaces the concatenation with feature-wise linear modulation (FiLM; Perez et al., 2018), as the winning team had (Van Santvliet et al., 2025): a small network, initialised to the identity, maps age and sex to a scale and a shift of the pooled feature. Step 5 adds stage-2 fine-tuning. Step 5b removes age and sex altogether and step 5c lowers the upper band-pass frequency from 100 to 45 Hz. Steps 0 to 5c were single runs on a fixed set of 100,000 training records. B0, the end of the ladder, is step 5c with the negative cap applied inside the training script, as the organisers would train it, with three seeds. On B0 we also tried stage-2 learning rates of 5×10^{-6} and 10^{-6}, the latter the rate the authors of ECG-FM used. In rounds 1 and 2, 10% of the training patients (CODE-15%) or records (PTB-XL, SaMi-Trop) were held out for validation.

Round 2 tried to take the length cue away. *Fixed 5 s windows.* ECG-FM was pretrained on 5 s segments (McKeen et al., 2025), and with 5 s windows no record of at least 5 s is padded, so the network cannot observe the record length. Training draws one random 5 s window per record per epoch; validation and testing average the logits of $K = \lceil T/5\,\mathrm{s} \rceil$ evenly spaced windows. Another Challenge team also trained on random 5 s crops and averaged over windows at inference, taking the top quarter of the window probabilities (Dong et al., 2025).

There is also a mechanical reason to dislike padding. ECG-FM's front end opens with a group normalisation over time, one group per channel, computed over padded positions as well as real ones, so trailing zeros shift the features at every valid position even when attention masks them (the repository's tests show this for a 7 s record padded to 8 s and to 10 s). Pad every record to 10 s and the shift becomes a fixed function of record length that a head can read off. We would have wanted 5 s windows for this reason alone.

Length-matched negatives. When negatives are subsampled, they are drawn so that their distribution over three length groups (shorter than 7.5 s, 7.5 to 9.9 s, at least 9.9 s) equals that of the positives, and record length then carries no information about the label in training or validation. This is frequency matching of controls to cases on a confounder, the group-balancing remedy for spurious correlations (Idrissi et al., 2022; Sagawa et al., 2020); it costs nothing here, since the cap discards two thirds of the negatives anyway. *Weight interpolation.* A frozen encoder beat a fine-tuned one on hidden data for one Challenge team (Wang et al., 2025), and fine-tuning

Table 2. The three rounds and their decision rules. Each rule was written down, with a date, before the runs it governed; supplementary section S1 gives the full texts. SE: standard error of a paired bootstrap difference.

Round	Candidates	Primary view and metric	Adoption rule (short)	Outcome
1, ablation	Steps 0–5c (single runs), then B0 (three seeds)	Pooled TPR@5% (view A)	One change per step; band-pass cut-off and negative cap chosen by written rules	B0: step 5c with the negative cap
1, learning rate	Stage-2 rates 2×10^{-5}, 5×10^{-6}, 10^{-6}	Pooled TPR@5% (view A)	A lower rate must win by more than 1 SE without losing AUROC	None; 2×10^{-5} kept
2	M1, W1, MW at $\alpha = 0$, 0.5, 1	S7 TPR@5%	Beat B0 on S7 TPR@5% by more than 1 SE (or S7 AUROC by more than 0.010); S10+ AUROC not lower by more than 1 SE; length-cue gap must shrink	None
3, fixed protocol	B0′, W1′, W1R, W1S, overlapping windows, five W1′ seeds	S10+ AUROC; S7 TPR@5% as guard	Beat W1′ by more than 1 SE on S10+ AUROC, or on S7 TPR@5% without losing 1 SE of S10+ AUROC	W1R, three-seed ensemble: the entry

can degrade out-of-distribution performance (Kumar et al., 2022), so we interpolated the weights of the stage-1 model (θ_1, frozen encoder) and the stage-2 model (θ_2) as $(1-\alpha)\theta_1 + \alpha\theta_2$ (Wortsman et al., 2022) and scored $\alpha = 0$, 0.5 and 1. *Seed ensembles* average three models that differ only in the random seed (Lakshminarayanan et al., 2017), as in earlier Chagas models and one Challenge entry (Jidling et al., 2023; Nejedly et al., 2025). Round 2 trained matched negatives (M1), fixed windows (W1) and both (MW) from B0, three seeds each.

Round 3 fixed the validation protocol. Rounds 1 and 2 had selected checkpoints on the pooled validation TPR@5%, which the length cue inflates, and had carved the validation set after the negative cap using the training seed, so each seed validated on different records. In round 3 the validation set is drawn before the cap, at natural prevalence, from 10% of the training patients stratified by source, label and length group, with a fixed seed, and PTB-XL is grouped by patient when its metadata table is available. It has 29,250 records and 681 positives and is the same for every run. Checkpoints are selected on the validation TPR@5% within 7.34 s CODE-15% records, which hold 534 of its positives (84 are 10 s records). The negative cap becomes a training prevalence of 6.5%, the value the 100,000-record cap gave on our split, so that it transfers to the organisers' larger training set.

Under this protocol we re-trained the baseline and the fixed-window configuration with three seeds. Because these are new models we call them B0′ and W1′. W1 was carried forward instead of MW because the two did not differ on S10+ AUROC by more than one SE in round 2 and W1 is the simpler change with the higher S7 score; length-matched sampling is therefore not part of the entry. We tested three additions to W1′. W1R adds the hard-negative ranking loss of Hong et al. (2025): a margin-ranking loss on the top 20% of negatives per batch by predicted probability, with margin 0.5 scaled by the difference of the soft targets and weight 2.5 relative to the cross-entropy. W1S drops PTB-XL, which exists only as negatives and is absent from the hidden data. The third infers over 5 s windows every 2.5 s, without retraining. Two further W1′ seeds tested a five-seed ensemble. The hidden cohorts are 10 s recordings, so round 3 made the S10+ AUROC its primary metric and kept the S7 TPR@5% as a guard.

3 Results

3.1 The entry, re-created, and a model that reads the record length

Step 0 of the ladder re-creates our Challenge entry in the new code (supplementary section S6 lists how it differs from the shipped model). It scored 0.412 in TPR@5% on view A and 0.377 within 7.34 s records (table 3).

Every model we trained, step 0 included, gave short records higher probabilities whatever their label. For B0 the mean probability of held-out CODE-15% negatives was 0.57 for short records and 0.40 for long ones, a length-cue gap of +0.17 (figure 1(b)); for step 0 the gap was +0.30. We noticed late, after round 1 had already chosen a model to ship. A test written when we rebuilt the pipeline had shown that the front end turns trailing zeros into a shift that depends on record length, and we noted then that length varied with the data source. Nine days later, when we set the record lengths against the labels, we saw that it varied with the label too. Every comparison below is made within length groups.

Table 3. The ablation ladder, scored on the pooled views and within record-length groups. Steps 0 to 5c are single runs; B0 and the uncapped run are means over seeds.

		Pooled (A)		Pooled (B)	Within 7.34 s (S7)		10.24 s (S10)	Length-cue
	Configuration	TPR@5%	AUROC	AUROC	TPR@5%	AUROC	AUROC	gap
0	Original entry (frozen encoder)	0.412	0.847	0.851	0.377	0.807	0.720	+0.30
1	+ pipeline fixes, validation split	0.442	0.862	0.857	0.376	0.812	0.733	+0.30
2	+ soft labels, class weighting	0.427	0.856	0.850	0.365	0.802	0.711	+0.16
3	+ augmentation	0.407	0.837	0.836	0.360	0.785	0.700	+0.17
4	+ FiLM demographics	0.434	0.852	0.850	0.373	0.807	0.705	+0.14
5	+ two-stage fine-tuning	0.435	0.854	0.850	0.378	0.806	0.727	+0.13
5b	demographics removed	0.438	0.848	0.849	0.399	0.800	0.739	+0.13
5c	0.5–45 Hz band-pass	0.437	0.842	0.847	0.400	0.799	0.733	+0.12
B0	5c, negatives capped in training (3 seeds)	0.436	0.829	0.830	0.384	0.787	0.695	+0.17
	all negatives, no cap (2 seeds)	0.427	0.791	0.792	0.382	0.751	0.632	+0.07

Table 4. Stage 2 against stage 1 (round 2, three seeds). "Both remedies" is fixed 5 s windows with length-matched negatives; α interpolates the weights between the stage-1 model of each run ($\alpha = 0$, encoder frozen) and its fine-tuned model ($\alpha = 1$). Δ is the difference from the baseline mean, with the SE of a paired bootstrap. Supplementary section S4 gives all of round 2.

	Pooled	Within 7.34 s records (S7)				10 s format (S10+)		Length-
Configuration	TPR@5%	TPR@5%	Δ	AUROC	Δ	AUROC	Δ	cue gap
Baseline (B0), fine-tuned	0.436	0.384	–	0.787	–	0.779	–	+0.17
Both remedies, stage 1 only ($\alpha = 0$)	0.354	0.343	-0.041 ± 0.009	0.775	-0.012 ± 0.006	0.797	$+0.017 \pm 0.017$	+0.00
Both remedies, halfway ($\alpha = 0.5$)	0.383	0.370	-0.014 ± 0.007	0.798	$+0.011 \pm 0.004$	0.833	$+0.054 \pm 0.012$	+0.01
Both remedies, stage 2 ($\alpha = 1$)	0.403	0.375	-0.009 ± 0.005	0.795	$+0.008 \pm 0.003$	0.841	$+0.062 \pm 0.011$	+0.01

3.2 The ladder, within length groups

Within 7.34 s records the ladder is almost flat. Each step was a single run and the bootstrap SE of TPR@5% in that group is about 0.013, so most differences between neighbouring rows are noise. The largest single step, oddly, was a removal: dropping age and sex (step 5 to 5b) added +0.021 to the S7 TPR@5%, about 1.6 SE, and no step moved it by two. Ranked by pooled score the ladder tells a different story, with the pipeline fixes of step 1 on top; ranked within 7.34 s records, the steps without age and sex lead. Which ordering one believes depends on whether one lets the model read the record length.

Two steps did more to the cues than to the score. Soft labels and class weighting halved the length-cue gap (+0.30 at step 1, +0.16 at step 2), and augmentation raised the mean probability of PTB-XL records from 0.08 to 0.32, probably because resampling every record to and from other rates hides the 500 Hz sampling that had marked PTB-XL as a source of negatives.

The demographic input was doing something we had not intended. FiLM, added at step 4, raised the pooled score and pulled the mean probability of PTB-XL records down to 0.24; within 7.34 s records it moved the score by about one standard error, which at this sample size is noise. Then, once the encoder was fine-tuned with age and sex still fed in, the PTB-XL mean collapsed to 0.01. PTB-XL is an all-negative European cohort with its own age and sex distribution, and 0.01 is the mark of a model that has learned to recognise the cohort, not the disease. Taking age and sex out brought the value back to 0.07 and cost nothing within length groups. The final model has none.

3.3 Fine-tuning ECG-FM

With the cue removed, stage 2 did earn its keep. In the round-2 configuration with both remedies the stage-1 model ($\alpha = 0$, encoder frozen) scored 0.343 within 7.34 s records and the fine-tuned model ($\alpha = 1$) 0.375, all three seeds moving the same way, and the AUROC followed: 0.775 and 0.795 within 7.34 s records, 0.797 and 0.841 on the 10 s format (table 4). Interpolating halfway ($\alpha = 0.5$) gave 0.370, nothing we would trade full fine-tuning for.

We should be plain about what the frozen arm is: the stage-1 model of the same run, a head trained for 2,000 steps, not a frozen encoder trained until it stopped improving. Step 0, a frozen encoder with a head trained for 16 epochs, is nearer the latter — it differs from B0 in other ways too — and within 7.34 s records it scored 0.377 against 0.384 for the fine-tuned B0, a difference we cannot tell from noise. So the honest statement is narrower than the one we would like to make. Stage 2 improves on stage 1; whether a fine-tuned ECG-FM beats a patiently trained frozen one, our data do not say.

Stage 2 was over almost before it began. At the base rate every run peaked in validation at the first or second epoch and then slid (supplementary section S2). We assumed this meant the rate was too high and that a gentler one would let the encoder settle somewhere better. It did not. At 10^{-6}, the rate the ECG-FM authors used, the peak came later (epoch 6.7 on average, against 1.7)

Table 5. Round 3, under the fixed validation protocol (three seeds unless stated). B0′ and W1′ are re-trained and are not the round-2 models. Δ is the difference from the B0′ mean, with the SE of a paired bootstrap. Supplementary section S4 shows the same runs seed by seed.

	Pooled		Within 7.34 s records (S7)			10 s format (S10+) AUROC			Length-	Source
Configuration	TPR@5%	TPR@5%	Δ	AUROC	Δ	mean	Δ	ensemble	cue gap	gap
Re-trained baseline (B0′)	0.427	0.385	–	0.788	–	0.781	–	0.809	+0.15	−0.04
+ 5 s windows (W1′)	0.422	0.383	-0.002 ± 0.005	0.794	$+0.006 \pm 0.002$	0.824	$+0.043 \pm 0.010$	0.834	+0.08	−0.02
W1′ with 5 seeds	0.417	0.375	-0.010 ± 0.005	0.792	$+0.004 \pm 0.002$	0.824	$+0.043 \pm 0.010$	0.837	+0.08	−0.02
W1′ + ranking loss (W1R)	0.425	0.389	$+0.004 \pm 0.005$	0.797	$+0.009 \pm 0.003$	0.827	$+0.046 \pm 0.012$	0.841	+0.01	−0.01
W1′ without PTB-XL (W1S)	0.414	0.384	-0.001 ± 0.005	0.793	$+0.005 \pm 0.003$	0.819	$+0.038 \pm 0.011$	0.839	+0.07	−0.01
W1′, windows every 2.5 s	0.422	0.383	-0.002 ± 0.005	0.794	$+0.006 \pm 0.002$	0.822	$+0.041 \pm 0.010$	0.832	+0.09	−0.02

but no higher, and the test score fell as the rate fell: 0.362 within 7.34 s records, 0.384 at the base rate. We had expected the opposite, and kept 2×10^{-5}. More data did not help either. Training on every negative instead of the capped third lowered the AUROC within short records to 0.751 (B0: 0.787), and validating every 500 steps, so as not to miss the early peak on the larger set, did not recover it; every later configuration kept the cap.

With the encoder frozen, the head weighted the middle layers most (layer 7 carried 3.2 times the weight of layer 12), as Nguyen et al. (2025b) found for another ECG transformer; fine-tuned, the weights stayed near uniform (supplementary section S3). We would like to read the flat profile as the fine-tuned encoder making every layer useful, but we cannot. The weights start uniform, they received a fraction of the optimiser steps step 0 gave them, and the runs that stayed longest in stage 2 drifted furthest from uniform. The flatness may be nothing more than a short stage 2.

3.4 Removing the cue

Both remedies narrowed the gap, matched negatives from +0.17 to +0.04 and the two together to +0.01, but the number that stayed with us is the one for fixed windows alone: +0.10. Given 5 s of unpadded signal the network has no way of knowing how long the record was, and it still scored short negatives higher. Something in the signal itself distinguishes the two formats. Our guess is the recording device.

Removing the cue cost us the pooled score, as it should have: TPR@5% on view A fell to 0.401 with matched negatives and 0.403 with both, from 0.436 for B0. Within 7.34 s records it bought nothing in TPR@5% (the largest change was -0.009 ± 0.005, with both), though fixed windows lifted the AUROC there by $+0.008 \pm 0.002$. The 10 s format is where the windows paid off. Fixed windows raised the S10+ AUROC by $+0.056 \pm 0.010$; matched negatives, which leave the input untouched, gained $+0.037 \pm 0.009$, and the two together $+0.062 \pm 0.011$. On S10 alone the AUROC was 0.695 for B0 and 0.773 with both remedies. By the rule we had written down beforehand, none of this was enough to adopt anything.

The fixed protocol of round 3 changed little by itself: B0′ scored 0.385 within short records and 0.781 in S10+ AUROC, close to B0 in round 2 (table 5). W1′ again left S7 where it was (-0.002 ± 0.005) and raised the S10+ AUROC by $+0.043 \pm 0.010$, and this time, with the 10 s format primary, that met the rule. All three additions passed against B0′, but none beat W1′ on S10+ by more than one SE. Overlapping windows changed nothing, and dropping PTB-XL did not make the positives more separable by source. On the S7 score the ranking loss gained $+0.006 \pm 0.005$ over W1′ without losing on S10+, which qualified it under the second clause. Applied as written, the rule made W1R the entry, on a margin of 1.3 standard errors that would not convince us by itself; what persuaded us was the gap. With fixed windows alone W1′ kept a length-cue gap of +0.08; with the ranking loss it was +0.01. On S10, which has no SaMi-Trop records, the AUROC rose from 0.675 for B0′ to 0.759 for W1R.

3.5 Seed ensembles

It is a little deflating to report that the most dependable gain in the study came from averaging three seeds. The baseline's TPR@5% within short records went from 0.384 to 0.404 and its AUROC there from 0.787 to 0.801, about as much as the largest step of the ladder, and without resting on a single run. The ensembles also carried the fixed-window advantage on the 10 s format: 0.857 and 0.854 in AUROC for fixed windows and the combined configuration, against 0.801 for the baseline ensemble. The W1R ensemble reached 0.410 on S7, the highest S7 score of any ensemble in rounds 1 to 3, and 0.841 in S10+ AUROC, in line with the round-2 fixed-window ensembles. Five W1′ seeds gained $+0.003 \pm 0.003$ over three on S10+, which gave us no reason to train more members.

The post-Challenge entry is the three members of W1R trained inside a single call of the Challenge training script. It scored 0.416 on S7 (AUROC 0.808) and 0.854 in AUROC on S10+, within the confidence interval of the round-3 W1R ensemble.

Table 6. Scores on the hidden data, as returned by the organisers, for our 2025 entry (Nguyen et al., 2025a) and the post-Challenge entry (W1R, three seeds). The test mean, over the three test cohorts, ranks entries in the Challenge. No confidence intervals are returned, and AUROC only for the post-Challenge entry.

	Original entry (2025)	Post-Challenge entry		Difference
Cohort	TPR@5%	TPR@5%	AUROC	in TPR@5%
Hidden validation	0.323	0.477	0.794	+0.154
REDS-II (test)	0.329	0.371	0.745	+0.042
SaMi-Trop 3 (test)	0.281	0.371	0.780	+0.090
ELSA-Brasil (test)	0.125	0.125	0.582	0.000
Test mean	0.245	0.289	–	+0.044

3.6 The hidden cohorts

The organisers ran the training and inference scripts of the post-Challenge entry (commit 355ff92, submission F28 of the focus issue) and scored it on the hidden cohorts on 19 October 2026. Table 6 sets it beside our original entry, scored on the same cohorts in 2025.

The test mean rose by +0.044, all of it on two cohorts: REDS-II gained +0.042 and SaMi-Trop 3 +0.090. ELSA-Brasil is the cohort we understand least. Both our entries scored 0.125 there, and so did the winning entry (PhysioNet, 2025); the score seems to take only a handful of distinct values on that cohort, and we cannot tell whether our model failed there or whether the metric has simply run out of resolution.

For context, the highest official test mean was 0.323 (REDS-II 0.468, SaMi-Trop 3 0.376, ELSA-Brasil 0.125), the second 0.283 and the median 0.217 (Reyna et al., 2025; PhysioNet, 2025). A score of 0.289 would sit between the first and the second, but an entry submitted after the Challenge closed is not ranked, and it was built with the published methods of other teams.

4 Discussion

4.1 Fine-tuning ECG-FM

During the Challenge we told ourselves a tidy story about fine-tuning overwriting what pretraining had taught the encoder. We no longer trust the evidence for it. It was one pooled comparison, with single-stage fine-tuning, on data where every model we have trained since reads the record length, and we never re-ran it. Measured within recording length, a short second stage helped, in the direction the winning team and the authors of ECG-FM had found (Van Santvliet et al., 2025; McKeen et al., 2025), and it helped early. We were wrong about the learning rate and about the data: a gentler rate did not help, and training on every negative was worse than training on a third of them. If we were starting again on a task like this one, we would run stage 2 at a base rate near 2×10^{-5}, not the 10^{-6} its authors used, validate often enough to catch a peak inside the first two epochs, and cap the negatives. Why a third of the negatives beats all of them is the result in this paper we would least like to defend and most like to understand; validating more often on the full set, which is what a missed early peak would call for, did not close the gap.

The input and the head told us less about the encoder than we had hoped. Fixed 5 s windows do three things at once: they give ECG-FM the segment length it was pretrained on, they remove the padding that the group normalisation turns into a length signature, and they hide the record length altogether. We cannot apportion the gain among the three. We do know that matched negatives, which touch none of the input, also gained on the 10 s format, so some of what the windows bought was simply the cue leaving. We did not add the ranking loss to remove the cue. We added it because it acts on the top of the ranking, which is all the Challenge score measures, and because Hong et al. (2025) had used it; it turned out to finish the job the windows had started. Our guess at the mechanism is that, under the cue, the negatives scored highest in a batch are disproportionately short records, so a loss that pushes them below the positives pushes the format signal out with them — but we never tabulated the lengths of the negatives it selected. Reading all 12 layers costs twelve scalar weights, and we would keep it, although we never tested it against the last layer alone. Age and sex we would leave out. With fine-tuning they let the model recognise PTB-XL records without finding more Chagas, and the winning team's use of FiLM (Van Santvliet et al., 2025) need not contradict that: in our ladder FiLM helped the pooled score only while the encoder was frozen, and a frozen or differently trained encoder might still gain from it.

4.2 The shortcut, and what it means for anyone using CODE-15%

We do not know why record length predicts the label. The two lengths probably correspond to two recording systems or software versions in the telehealth network that collected CODE-15% (Ribeiro

et al., 2020), deployed in regions or periods with different prevalence of Chagas disease, which would also explain why a model that cannot see the length still partly separates the formats. The public data carry no device or site information, so we could not test this. The cue probably accounts for only a few hundredths of the gap between internal and hidden scores that almost every team saw; it raises step 0's AUROC on CODE-15% records by about 0.044 (supplementary section S5), and the rest belongs to cohort, label quality and device. Two of the three highest-ranked entries used augmentations that blur it. The winning team cropped records at random to between 5.65 and 10 s and zero-padded them (Van Santvliet et al., 2025), and the third-placed team masked the beginning and end of records "to simulate shorter signals" (Hong et al., 2025). Neither team described the association with the label. In our own Challenge paper we listed random padding or windowing as future work (Nguyen et al., 2025a).

For anyone working with CODE-15%, the practical point is that a pooled score cannot rank models. Ours rose by 0.023 over the ladder while the score within 7.34 s records stood still, and the patient-grouped split at natural prevalence, both of which we consider necessary, did nothing to prevent it. Scoring 7.34 s and 10.24 s records separately costs nothing; the duration-only baseline and the length-cue gap are cheap checks; and pooled scores should be expected to fall when the cue goes. Length-matched negatives are free when negatives are plentiful, and a window that never needs padding hides the length, but neither guarantees the cue is gone, as our fixed windows showed. A pretrained encoder also deserves its pretraining format, here 5 s at 500 Hz, and a look at what its front end does with padding. Had we known about the length association at the start, we would have scored within format from the first run, and we would not have spent a round ranking changes by a pooled score that the cue was moving.

4.3 One external score, and what it cannot show

We have one scored entry to go on. It shows that the revised pipeline, trained by the organisers with the unmodified entry scripts, found more positives among its top 5% than our original entry on two of the three hidden test cohorts, in the direction the within-format evidence predicted. Every hidden AUROC was below the 0.854 the entry reached on the 10 s format of our held-out set, the lowest being 0.582 on ELSA-Brasil. That was to be expected: the held-out set chose the entry, and the hidden cohorts are a different population with serological labels and their own devices. An earlier network reached 0.68 and 0.59 on its own versions of REDS-II and ELSA-Brasil (Jidling et al., 2023).

One number cannot apportion credit among the changes that separate the two entries. Between the original and the revised entry sit the pipeline corrections, soft labels, augmentation, the removal of age and sex, fine-tuning, fixed windows, the ranking loss, the validation protocol and the ensemble, and the hidden score tests the lot as a package. We would dearly like to know whether fine-tuning alone would have moved REDS-II, or whether dropping age and sex mattered there; the only way to find out is a second scored entry that differs by one change. Without confidence intervals from the organisers we cannot say how much of the difference is sampling variation either. For scale, the bootstrap SE of TPR@5% on our 7.34 s held-out records was about 0.013.

The limitation that matters most is that we have one external score and it tests everything at once. Internally, the 10 s format of our held-out set is thin: 77 CODE-15% positives, and most of the positives in S10+ are SaMi-Trop records, which a model may recognise by source rather than by disease, although the round-3 gain also appeared on S10 alone. The held-out split was the selection set for every decision in three rounds, so the entry's internal scores are optimistic, and the round-3 rule, though fixed before its runs, was written after round 2 had shown the 10 s-format gain. Length-matched sampling was not re-tested under the fixed protocol. Everything else is the ordinary smallness of one study: one split, one backbone, no comparator without a foundation-model encoder, single runs on the ladder before B0, and self-reported labels in the largest source.

5 Conclusion

We went back to a frozen-encoder entry to ask how much more ECG-FM had to give, and spent most of the study learning how to measure the answer. A short second stage of fine-tuning at a moderate learning rate, unpadded windows of the length the encoder was pretrained on, a ranking loss and an ensemble of three seeds made an entry that the organisers scored higher than our original one, and we cannot yet say which of them did the work. On CODE-15% the comparison has to be made within recording length. Otherwise the length of the recording decides it.

Acknowledgments
The authors thank the organisers of the George B. Moody PhysioNet Challenge 2025, who also trained the post-Challenge entry and scored it on the hidden data, and the teams whose published methods informed this study. An AI coding assistant (Claude, Anthropic) was used under the authors' direction to implement and run the experiments, analyse the results and draft this manuscript. The authors reviewed the code, results and text and take responsibility for the content.

Funding
To be completed by the authors.

Author contributions
To be completed by the authors.

Data availability
The CODE-15%, SaMi-Trop and PTB-XL data are publicly available through the Challenge (Reyna et al., 2025). The code, the configuration of every run, the record-length table and the scripts that generate each figure, table and number in this paper are in the project repository (to be made public by the authors); the organisers' scores of both entries are in `EXTERNAL_SCORES.csv`, and the learned layer weights of every checkpoint are in `experiments/results/entry_final_r3/layer_weights.csv`. The decision rule of each round and the reason for every change of plan are recorded, with dates, in the repository's `DECISIONS.md`, `PLAN.md`, `PLAN_R2.md` and `PLAN_R3.md`.

Supplementary data
Supplementary material accompanies this paper: the full decision rules (S1), the stage-2 learning-rate sweep (S2), the learned layer weights (S3), rounds 2 and 3 in full (S4), the ablation AUROCs by length group (S5), reconstruction and timing details (S6) and views B and C (S7).

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